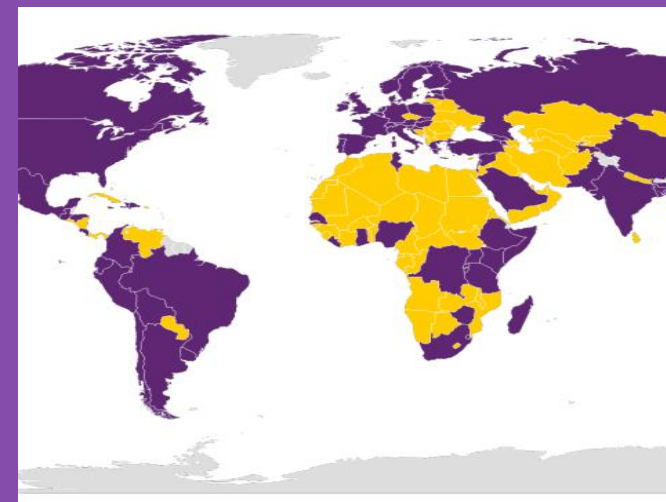


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Building the Foundations for AI in Tax Administration

Tax for Growth

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Why AI Matters for Development

- How can AI build tax capacity?

Tax capacity is a tax authority's ability to collect taxes, at reasonable administrative cost, with limited economic distortions and low distributional impacts.

(Jensen et al., 2024)

Why is Tax Administration ideal for AI deployment?

- Extremely rich administrative data
- Massive operational scale – decisions regarding audits and fraud could impact millions of taxpayers
- **76% of CATA members** believe AI will fully transform their functions
- Requires a shift in how information is managed
- But there exists a large gap between the hype of AI and raising revenue

The Engine and Application Framework

INTERNAL CAPACITY

The engine

Building the foundations required to operate



1. Data Plumbing

Digitise transactions, integrate registries, unique IDs.



2. Human in the loop

In-house data units, domain experts and data scientists.



3. Governance rules

Privacy, ethics and security frameworks.

enables

EXTERNAL IMPACT

The application

Delivering results for revenue and taxpayers



1. Enforcement

Fraud detection, risk scoring, customs valuation.



2. Compliance

Pre-filled returns, chatbots, lower compliance costs.



3. Governance rules

Microsimulation, scenario modelling, revenue forecasting.

The virtuous feedback loop: revenue gains and new data from external application are reinvested to upgrade internal capacity

1. Data Plumbing

- AI, ML, LLM and big data models operate by training on large quantities of data.
- High quality data is a challenge in many countries
 - a. Prioritise the digitisation of transactions.**
 - b. Integrate data siloes.**
 - c. Establish a “single reference point”**



What is Quality Data?



2. The Human in the loop

- **You need to get the right skills:** Without data analysis and engineering skills, the tax authority cannot respond to changes in tax codes, technology.
- Tax authorities face a clear **trade-off**: building in-house creates ownership, but it requires sustained investment in staff and capability.
- In our survey, 27% of respondents already had a specialised AI or big data team and another 33% have plans to develop such teams.
- **The whole tax administration needs to have a strong basis for using digital data management tools.**



3. Governance Rules

- Public trust is essential for tax administration. A lack of clear governance can lead to privacy breaches or algorithmic bias that erodes voluntary compliance.
- This framework rests on three core areas
 - Privacy and ownership
 - Data security
 - Data use and transparency

Digital Maturity



Level 1 Initial

Ad hoc and manual

Dimension 1. Scope

Isolated. Paper-based processes and ad hoc spreadsheets

Dimension 2. Data quality

Fragmented. Inconsistent identifiers; high manual error rates.

Dimension 3. Type of use

None. Manual checks or simple scripts.

Priority action

Digitise transactions (EDFs) and fix 'data plumbing'



Level 2 Controlled

Digitised and descriptive

Dimension 1. Scope

Localised. Specific departments (eg, VAT) use basic digital tools

Dimension 2. Data quality

Digitised. Key registries exist but remain siloed.

Dimension 3. Type of use

Descriptive. Dashboards asking "What happened?"

Priority action

Establish data stewards and launch simple ML pilots.



Level 3 Proactive

Integrated and predictive

Dimension 1. Scope

Generalised. Analytics used across all core functions.

Dimension 2. Data quality

Integrated. 'Data warehouse' with 360° taxpayer view.

Dimension 3. Type of use

Predictive. ML models asking "What will happen?"

Priority action

Formalise governance and operationalise ML workflows.



Level 4 Optimised

AI-native and adaptive

Dimension 1. Scope

Enterprise-wide. Fully embedded in all workflows.

Dimension 2. Data quality

Real-time. Continuous streams and unstructured data.

Dimension 3. Type of use

Prescriptive. AI optimising "How do we respond?"

Priority action

Implement MLOps, continuous retraining and audits.

What's in the Toolkit?

- A practical manual:
 - A self-assessment checklist for administrations to assess their level of digital maturity
 - A step-by-step roadmap to move from L1→L2 or L2→L3
 - Detailed use cases and country case studies





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